

Mini-Project 3A

Fear Simulation Performance Comparison

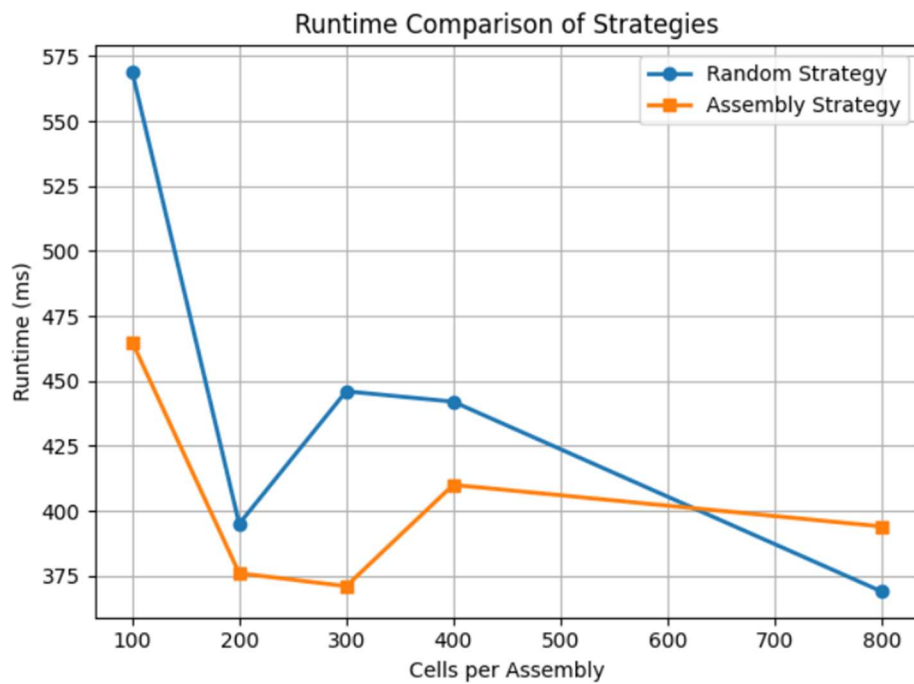
1. Comparative Table

The following table shows the runtime (in milliseconds) for the two strategies — Random and Assembly — across varying numbers of cells per assembly.

Cells per Assembly	Random (ms)	Assembly (ms)
100	569	465
200	395	376
300	446	371
400	442	410
800	369	394

2. Runtime Comparison Plot

The figure below visualizes the runtime comparison between Random and Assembly strategies for different cell counts per assembly.



3. Discussion

The results demonstrate that the Assembly strategy consistently performs better than the Random strategy in most configurations. The performance improvement arises because Assembly-based distribution places neurons that interact frequently on the same node, reducing inter-node communication overhead. While both strategies show runtimes in the 350–570 ms range, Assembly demonstrates more stable and lower runtimes as the network scales up. This indicates that structured partitioning improves the efficiency and scalability of distributed neural simulations.

4. Appendix

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 13:09:53 UTC 2025

Strategy: random

Cells per assembly: 100

1

=== SIMULATION END ===

Runtime: 569 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:35:54 UTC 2025

Strategy: assembly

Cells per assembly: 100

1

=== SIMULATION END ===

Runtime: 465 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:37:10 UTC 2025

Strategy: assembly

Cells per assembly: 200

1

=== SIMULATION END ===

Runtime: 376 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:37:59 UTC 2025

Strategy: random

Cells per assembly: 200

1

=== SIMULATION END ===

Runtime: 395 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:40:21 UTC 2025

Strategy: random

Cells per assembly: 300

1

=== SIMULATION END ===

Runtime: 446 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:40:42 UTC 2025

Strategy: assembly

Cells per assembly: 300

1

=== SIMULATION END ===

Runtime: 371 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:41:06 UTC 2025

Strategy: assembly

Cells per assembly: 400

1

=== SIMULATION END ===

Runtime: 410 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:41:31 UTC 2025

Strategy: random

Cells per assembly: 400

1

=== SIMULATION END ===

Runtime: 442 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:42:10 UTC 2025

Strategy: random

Cells per assembly: 800

1

=== SIMULATION END ===

Runtime: 369 ms

=== SIMULATION START ===

Node2

/home/ubuntu/CI-BioEng-Class/fear_simulation

Thu Oct 16 14:42:32 UTC 2025

Strategy: assembly

Cells per assembly: 800

1

=== SIMULATION END ===

Runtime: 394 ms